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**ADVANCED COLLEGE  
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# **Guptrix Note Management System**

## **Problem Statement:**

Students, professionals, and researchers often struggle to keep their notes organized, accessible, and manageable. Reliance on traditional paper notebooks or fragmented digital files frequently leads to data loss, duplication, and wasted time spent hunting for information.

While many existing solutions offer basic storage, they often lack the sophisticated organization and search capabilities necessary for today's information-heavy environment. This deficiency forces users to dig through endless folders, hindering productivity and complicating knowledge management. Furthermore, the lack of a centralized platform creates friction when accessing notes across multiple devices or locations, ultimately resulting in disorganized workflows and decreased efficiency. As digital information accumulates, there is an urgent need for a streamlined, web-based system that allows users to create, search, and manage notes effortlessly.

## **Proposed Solution:**

The Guptrix Note Management System is a web-based and app platform engineered to transform how you create, organize, and manage digital notes. It offers a secure, centralized hub, ensuring your information is accessible from anywhere with an internet connection.

With Guptrix, managing your knowledge base becomes intuitive. The system empowers you to create, edit, and update notes while organizing them into logical categories and collections. This structured approach drastically cuts down on retrieval time. Productivity is further boosted by a powerful search engine, allowing you to locate specific notes instantly via keywords, titles, or content queries, rather than manually browsing.

We prioritize your security and convenience. The platform features robust account management, ensuring that your personal information and sensitive notes remain private and accessible only to you. Built with a responsive interface that adapts to various devices, Guptrix helps you maintain a coherent, reliable, and user-friendly repository. By solving the fragmentation issues inherent in traditional note-taking, the system supports students, educators, and professionals alike in working smarter and more effectively.

# **FUNCTIONAL REQUIREMENTS**

Users:

## **R1: Registration\_User**

**Description:** Users should be able to create an account to access secure note management features.

**Input:** Username, email, password.

**Output:** Successfully registered account.

**Process:** The system validates user information, creates an account, and grants access to the platform.

## **R2: Login\_User**

**Description:** Users should be able to log in using valid credentials.

**Input:** Email, password.

**Output:** Successful login and user dashboard access.

**Process:** The system authenticates credentials and establishes a secure session.

## **R3: Create\_Note**

**Description:** Users should be able to create encrypted notes.

**Input:** Note title, note content.

**Output:** New note stored successfully.

**Process:** The system encrypts note content on the client side before storage.

## **R4: Edit\_Note**

**Description:** Users can modify existing notes.

**Input:** Note identifier, updated content.

**Output:** Note successfully updated.

Process: The system encrypts the updated content client-side before saving it.

#### **R5: Delete\_Note**

**Description:** Users can permanently remove notes.

**Input:** Note identifier.

**Output:** Permanent removal confirmation.

Process: The system permanently removes the note from storage and provides a confirmation.

#### **R6: Search\_Notes**

**Description:** Users can search their collection using keywords.

**Input:** Search keywords.

**Output:** Matching search results.

Process: The system scans note metadata to instantly display all matching results.

#### **R7: Upload\_Image**

**Description:** Users can attach image files to their notes.

**Input:** Image file.

**Output:** Image encrypted and attached.

Process: The system encrypts and stores images alongside the associated note data.

#### **R8: Draw\_And\_Save**

**Description:** Users can create sketches directly on a digital canvas.

**Input:** Canvas drawing data.

**Output:** Drawing saved securely.

Process: The system converts drawings into an encrypted format for secure saving.

#### **R9: Collaborate\_User**

**Description:** Users can initiate collaboration requests with others.

**Input:** Collaborator details, workspace identifier.

**Output:** Shared workspace access granted.

Process: The system manages permissions and syncs updates in real-time.

#### **R10: Cloud\_Synchronization**

**Description:** Securely synchronizes encrypted data with cloud storage.

**Input:** User note data.

**Output:** Consistent data across devices.

Process: The system syncs encrypted user data during active sessions.

#### **R11: Download\_Local\_Backup**

**Description:** Users can request an export of their notes.

**Input:** Export request.

**Output:** Downloadable encrypted JSON file.

Process: The system generates and provides a downloadable, encrypted backup file.

#### **R12: Import\_Local\_Backup**

**Description:** Users can restore their data from a backup file.

**Input:** JSON backup file.

**Output:** Restored user data.

Process: The system validates and decrypts the file before restoration.

#### **R13: Instant\_Lock\_System**

**Description:** Users can instantly secure their workspace.

**Input:** Lock command.

**Output:** Workspace locked.

Process: The system hides the active session behind a lock screen until re-authentication.

#### **R14: Session\_Management**

**Description:** Ensures security by managing user sessions.

**Input:** User credentials.

**Output:** Secure session validation/termination.

Process: The system creates, validates, and terminates sessions based on credentials.

#### **R15: Password\_Protected\_Notes**

**Description:** Users can apply secondary passwords to specific notes.

**Input:** Secondary password, note selection.

**Output:** Note secured with additional encryption layer.

Process: The system uses these credentials to add an additional layer of encryption.

#### **R16: Note\_Categorization**

**Description:** Users can organize their library by assigning category information.

**Input:** Category details, note selection.

**Output:** Note successfully categorized.

Process: The system associates category information with notes for management.

#### **R17: User\_Profile\_Management**

**Description:** Users can update their account details at any time.

**Input:** Updated profile information.

**Output:** Updated user records.

Process: The system validates profile changes and updates records accordingly.

### **R18: Zero\_Knowledge\_Storage**

**Description:** The system stores only encrypted user data to ensure privacy.

**Input:** User content.

**Output:** Private, encrypted storage.

Process: Encryption keys stay with the user so content is invisible to others.

## **NON-FUNCTIONAL REQUIREMENTS**

### **R1: Security**

**Description:** All notes and attachments must be encrypted before transmission and storage.

### **R2: Privacy**

**Description:** The system must implement a zero-knowledge architecture preventing server-side access to user content.

### **R3: Performance**

**Description:** Note creation, editing, and retrieval operations should complete within 2 seconds.

### **R4: Availability**

**Description:** The system should be available 24/7 except during scheduled maintenance.

### **R5: Reliability**

**Description:** The system should maintain 99.9% uptime with minimal service interruptions.

### **R6: Usability**

**Description:** New users should be able to learn the system within 10 minutes.

### **R7: Accessibility**

**Description:** The system should support accessibility standards and responsive design.

**R8: Data\_Backup\_Recovery**

**Description:** Users should be able to restore data from encrypted backup files.

**R9: Scalability**

**Description:** The system should support increasing numbers of users and notes without significant performance degradation.

**R10: Interoperability**

**Description:** The system should support operation across modern browsers and devices.

**R11: Maintainability**

**Description:** The software architecture should support future feature additions and updates.

**R12: Ad\_Free\_Operation**

**Description:** The platform should remain free from advertisements and third-party tracking systems.

**R13: Data\_Integrity**

**Description:** Stored encrypted notes must remain unchanged and protected from unauthorized modification.

**R14: Secure\_Cloud\_Storage**

**Description:** Cloud databases should only store encrypted user data and never plaintext note content.

**R15: Auditability**

**Description:** System activities such as login, logout, backup, and synchronization should be logged securely.



# CONSTRAINTS

## 1. **Client-Side Encryption Constraint**

The system must encrypt all note content on the user's device before transmission to cloud storage.

## 2. **Zero-Knowledge Architecture Constraint**

The server must not possess encryption keys capable of decrypting user notes.

## 3. **Internet Connectivity Constraint**

Cloud synchronization features require a stable internet connection.

## 4. **Browser Compatibility Constraint**

The system depends on modern web browsers that support JavaScript, Web Crypto API, and local storage.

## 5. **Storage Capacity Constraint**

The amount of data that can be stored may be limited by database and hosting resources.

## 6. **Performance Constraint**

Encryption, decryption, synchronization, and note retrieval operations must maintain acceptable response times.

## 7. **Device Resource Constraint**

Drawing tools, image storage, and large note collections may require sufficient memory and processing power.

## 8. **Cloud Service Dependency Constraint**

System availability depends on cloud hosting services and database infrastructure.

## 9. **Backup Management Constraint**

Users are responsible for securely storing downloaded encrypted JSON backup files.

## 10. **Password Recovery Constraint**

Due to zero-knowledge encryption, forgotten encryption passwords may prevent access to note contents.

#### **11. Collaboration Constraint**

Collaborative note sharing is limited to authenticated and authorized users.

#### **12. Security Constraint**

All communications between client and server must use secure encrypted protocols (HTTPS/TLS).

#### **13. Scalability Constraint**

System performance may be affected by a significant increase in concurrent users and stored notes.

#### **14. Data Integrity Constraint**

Imported backup files must follow the system's supported JSON structure and validation rules.

#### **15. Legal and Privacy Constraint**

The system must comply with applicable privacy and data protection regulations.

#### **16. Session Management Constraint**

User sessions must automatically expire after a predefined period of inactivity.

#### **17. Cross-Platform Constraint**

Features should function consistently across desktop, tablet, and mobile devices.

#### **18. Third-Party Service Constraint**

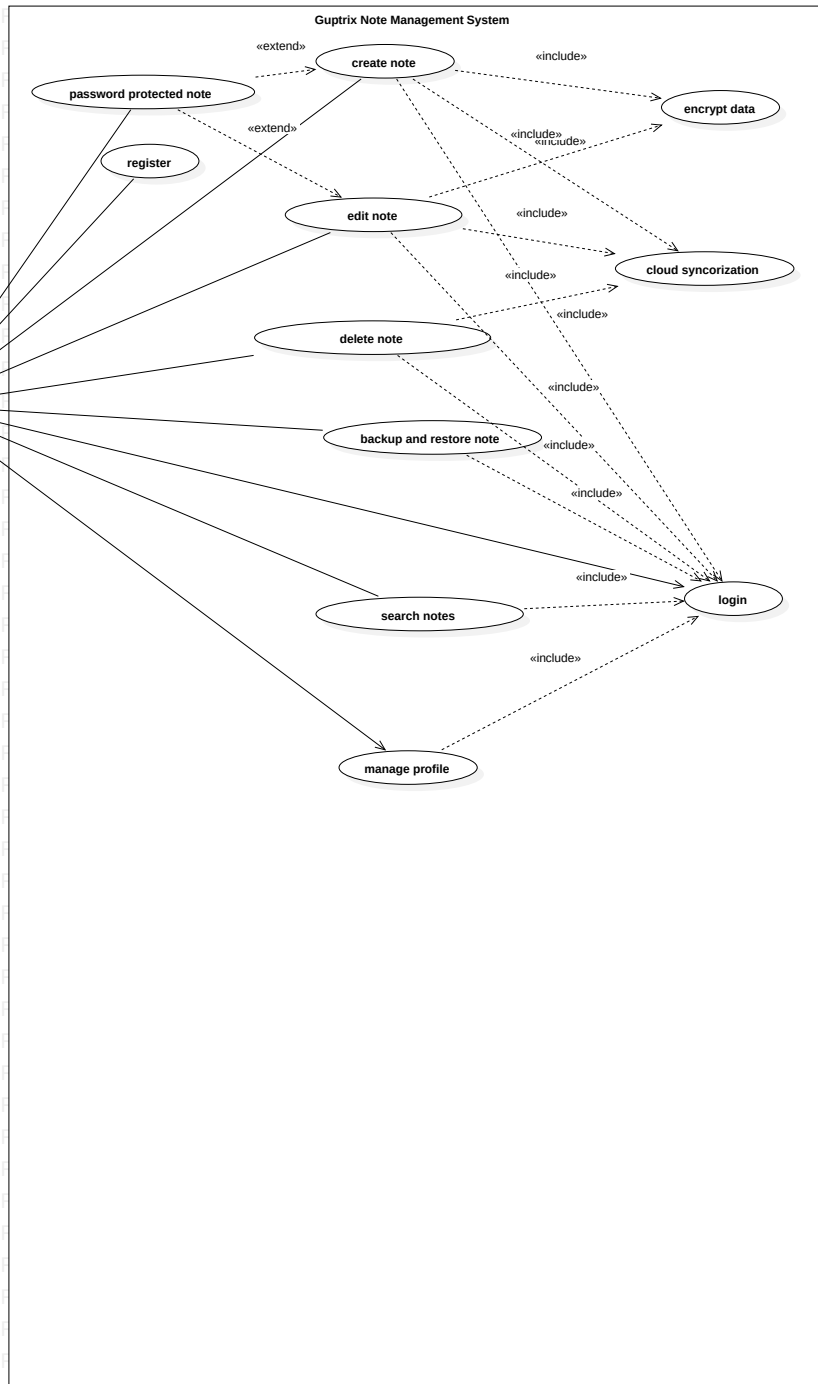
Email verification, authentication services, and cloud infrastructure depend on external service providers.

#### **19. Maintenance Constraint**

System updates and maintenance may temporarily affect service availability.

#### **20. Resource Constraint**

Project development is limited by available budget, development time, hosting resources, and personnel.



Model3::ClassDiagram1

